

ESP32 Multimodal NIDS — Status Update

2026.08.09 · matched-load · deployable `nids-counts`

Bottom line (now)

1. **Density contract fixed** — firmware `win_pkts` / `win_dens` ; offline windows match on-device counts.
2. **Matched-load collected & trained** — NORMAL includes busy (label 0) + DEAUTH / SYN / ARP (START/STOP).
3. **Multimodal research still stands** — HIDS matters; shallow trees turn it into a **stress proxy** (heap → then `udpfail`).
4. **Deploy export ≠ drop HIDS** — board runs **nids-counts** (deauth / packet counts; HIDS+RSSI neutralized at train time).
5. **Board smoke: IDLE stable · DEAUTH fires · SYN/ARP left weak on purpose** (not worth a smoke test).
6. **B' dual-expert offline done** (gated OR). **Ask:** lab → **AUTH_FLOOD**; full ARP features after meeting.

What we are building

Edge window IDS on ESP32 (not a per-packet firewall):

- **Sense:** promiscuous 802.11 + host counters
- **Decide:** shallow DT every **100 ms** → `model.h`
- **Act:** HIPS **off** (until FPR story is deployable)

Layer	Still in the project?	Role now
Data + HIDS fields	Yes	Collected; used in ablations / narrative
Offline full / RF / PCA / LDA	Yes	Analysis & advisor pack
On-device <code>nids_predict</code>	nids-counts	Stable WIDS-lite for demo / low FP

HIDS is not deleted — it is not the sole split for the MCU tree right now.

Pipeline this cycle

```
flash win_* firmware → matched-load capture  
→ aggregate [firmware] → balance PASS  
→ train + heap / export ablations  
→ board: IDLE + DEAUTH smoke
```

Capture	Role
20260809_000851	Matched-load train + board model
Density short busy 20260806_231142	Contract validation (done)

Failed / trial CSVs archived under `data/archive/` — not used for metrics.

Matched-load numbers (hold-out)

Dataset: 6354 windows · NORMAL 4947 · DEAUTH 656 · SYN 495 · ARP 256 · **balance PASS**

Export	F1	FPR	What the tree uses	Board IDLE
full multimodal	0.54	0.48	mostly <code>udpfail</code>	—
nids-only (no HIDS)	0.63	0.06	<code>rssiVar / snr</code>	FAIL (always yellow)
nids-counts	0.57	0.06	<code>deauth / pkts</code>	PASS

Per-attack recall (**nids-counts**): DEAUTH ~0.80 · SYN ~0.37 · ARP ~0.02

Matched-load cut heap dominance & busy FP vs old stress-proxy behavior; deploy tree trades L2 recall for stability.

Why neutralize HIDS for deploy?

Evidence	Takeaway
Full matched tree	Heap importance ↓, but <code>udpfail</code> takes over → high FPR
nids-only on device	RF stats don't transfer across environments
nids-counts on device	IDLE quiet; DEAUTH windows show high <code>deauth</code> / <code>tgt</code>

Research story: multimodal + HIDS ablation still required.

Product story: MCU ships a **stable subset** until dual-expert / better HIDS features exist.

Board smoke (nids-counts)

Scenario	Result
IDLE	Long stretches <code>attack=0</code> ; almost no inference spam
DEAUTH	<code>[INFERENCE] attack window</code> with large <code>deauth</code> / <code>tgt</code>
SYN / ARP	Skipped — offline already weak; won't change the Ask

Note: `wifi: This was attack!!! ... SA Query` is **ESP-IDF**, not `nids_predict` .

B' Dual-expert (offline — HIDS back as 2nd path)

Same matched dataset · Expert **W** = nids-counts · Expert **H** = HIDS-only · fusion offline

Model	F1	FPR	ARP	SYN	DEAUTH
W (deploy today)	0.57	0.06	0.02	0.37	0.80
H alone	0.60	0.35	1.00	0.92	0.94
WvH naive	0.57	0.40	1.00	0.94	0.96
WvH (H if $p \geq 0.85$)	0.63	0.09	0.02	0.49	0.96

- HIDS is **not abandoned** — it recovers stress-shaped attacks offline.
- Naive OR is not deployable; **gated OR** is the practical fusion.
- **ARP still needs new air features** (full B after the meeting).

Offline models (if asked)

- Pipeline: **PCA / LDA**, DT hyperparam grid, **RF** (+ optional boosters).
- Deeper DT offline F1 ~**0.72** vs MCU depth=4 ~**0.54**.
- Dual-expert script: `host/train/dual_expert_eval.py` → `docs/figures/dual_expert_20260809.json` .

Proposed next (ordered)

Order	Track	Status
1	B' dual-expert offline	Done (gated OR numbers above)
2	C AUTH_FLOOD lab day	Next experiment — handbook ready
3	Full B ARP/SYN air features	After 08/13 — firmware + recollect

Not next: blind matched re-collect · HIPS on · lock Git · MCU ensemble chase.

Ask

1. OK to run Phase C (`AUTH_FLOOD`) as the next lab capture?
2. OK to schedule full ARP feature work after this meeting (dual-expert does not fix ARP visibility)?

Our lean: keep MCU on nids-counts · present B' gated OR as HIDS path · lab → **AUTH** · then ARP features.

[Backup] Lab sketch

```
Kali  -- air --> ESP32 (STA + promiscuous)
Kali  -- START/STOP --> Collector (Pi / Mode P)
ESP32 -- syslog (win_*) --> Collector → windows → model.h
```

HIPS off while iterating. Hotspot + host-only label path as in runbook.

[Backup] Export variants (same `model.h` layout)

Flag	Neutralized at train	Intent
<code>full</code>	—	Multimodal DT (high FPR here)
<code>nids-only</code>	all HIDS	Wireless including RSSI
<code>nids-counts</code>	HIDS + rssi/snr	Deploy / board smoke
<code>no-heap</code>	heap only	Ablation

```
python host/train/analyze_and_train.py `
  --dataset data/windows/nids_windows_20260809_000851.csv `
  --export-variant nids-counts
```

[Backup] Feature groups

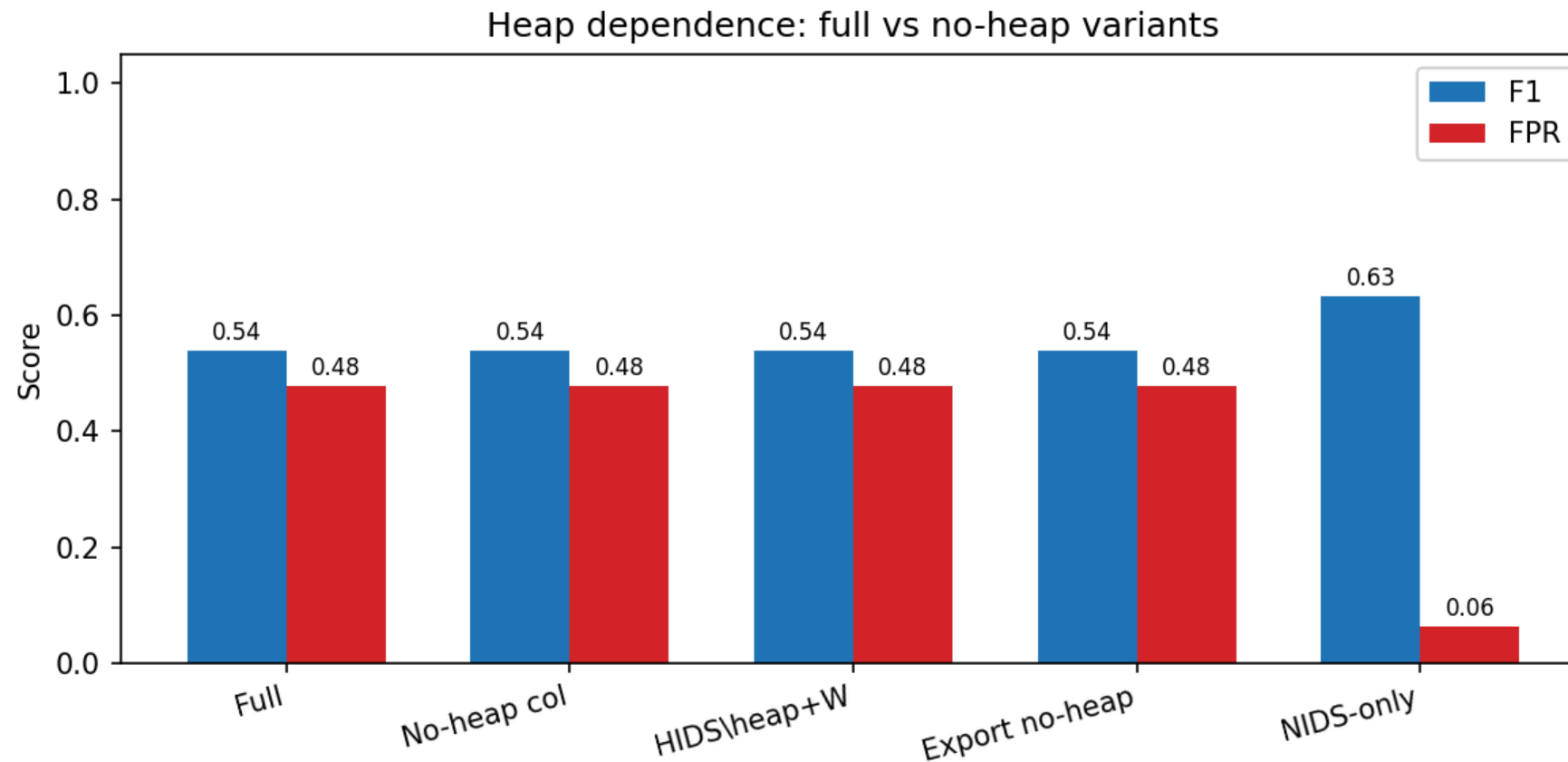
	Features
Counts / WIDS	total_packets , packet_density , beacon/deauth/probe/auth , deauth_targeted , seq_jump
RF soft	rss_i_mean , rss_i_var , snr_mean
HIDS	heap , minheap , reconn , qpeak , udpfail , backlog

nids-counts train: HIDS + RF soft held at NORMAL median → splits on deauth / counts.

[Backup] Artifacts

Path	Content
docs/0809report.md	This deck
docs/0806report.md	Historical Phase-A ask (superseded for meetings)
note/private/README.md	Index + short/mid/long plan
note/private/eval/matched_load_eval_20260809.md	Numbers + board notes
docs/figures/eval_metrics*.json	Metrics snapshots
main/model.h	Current nids-counts export

[Backup] Heap / export ablation (figure)



[Backup] Decision tree (current export)

Exported Decision Tree (DecisionTree)

